

Partial result for Kerr–Li Question 9.4

Product-type UHF automorphisms have CA entropy 0 or ∞

status: partial result, not a full solution

Source Question

The source is David Kerr and Hanfeng Li, *Dynamical entropy in Banach spaces*, arXiv:math/0407386. Question 9.4 appears after their proof that the tensor product shift on $M_p^{\otimes \mathbb{Z}}$ has infinite contractive approximation entropy:

Is there a $*$ -automorphism of $M_p^{\otimes \mathbb{Z}}$ (or any other simple AF algebra) with finite nonzero CA entropy?

$$\begin{aligned} &= \left| \sum_{k=1}^n \sigma_k(c_k u_k + d_k v_k) \right| \\ &= \sum_{k=1}^n (c_k^2 + d_k^2)^{1/2} \end{aligned}$$

so that $\left\| \sum_{k=1}^n c_k u_k + d_k v_k \right\| = \sum_{k=1}^n (c_k^2 + d_k^2)^{1/2}$, which establishes the proposition. $\square$

Theorem 9.3. The tensor product shift α on $M_p^{\otimes \mathbb{Z}}$ has infinite CA entropy.

Proof. By Lemma 9.2 every finite subset of the unit sphere of Y_1 satisfies the hypotheses of Lemma 9.1, and so we obtain the result. $\square$

Theorem 9.3 prompts the following question. (Compare the discussion after Theorem 7.4.)

Question 9.4. Is there a $*$ -automorphism of $M_p^{\otimes \mathbb{Z}}$ (or any other simple AF algebra) with finite nonzero CA entropy?

We also remark that we don't know whether there exists a $*$ -automorphism of a simple purely infinite C^* -algebra (in particular, the Cuntz algebra $\mathcal{O}_2$) with finite nonzero CA entropy (cf. [12]).

10. ISOMETRIC AUTOMORPHISMS OF ℓ_∞

By the Banach-Stone theorem, for every isometric automorphism α of ℓ_∞ there are a

Definitions and Source Facts

We use Kerr–Li's notation $h_c(\alpha)$ for the contractive approximation entropy of an isometric automorphism α of a Banach space.

Two source facts are used.

- Proposition 3.1: if Y is an α -invariant closed subspace, then $h_c(\alpha|_Y) \leq h_c(\alpha)$, and $h_c(\alpha)$ is computed as a supremum over increasing finite sets whose α -orbits are total.
- Lemma 9.1: if a finite set Ω of unit vectors satisfies $\|x + y\| < 2$ for distinct $x, y \in \Omega$, and

$$\left\| \sum_{k=0}^{n-1} \alpha^k(x_k) \right\| = n \quad (x_k \in \Omega, \ n \geq 1),$$

then $h_c(\alpha, \Omega) \geq \log |\Omega|$.

Idea of the Proof

For a product-type automorphism, the tensor-factor permutation is the only source of large orbit geometry. If all permutation orbits are finite, every finite set of algebra elements lives, up to approximation, inside a finite-dimensional invariant tensor block, so entropy is zero.

If there is an infinite tensor-factor orbit, a single matrix factor contains a real two-dimensional Euclidean sphere of self-adjoint unitaries. Iterating different choices from a finite subset of that sphere places them in distinct commuting tensor factors. Product states then make the norm of every such orbit sum exactly additive. Since the finite subset of the Euclidean sphere can be chosen with arbitrarily large cardinality, Kerr–Li’s Lemma 9.1 forces arbitrarily large entropy lower bounds, hence infinite entropy.

The Partial Result

Theorem 1. *Let J be a countably infinite set, let $p \geq 2$, and let*

$$A = \bigotimes_{j \in J} M_p$$

be the spatial infinite tensor product with respect to the units. Suppose $\alpha \in \text{Aut}(A)$ is product-type in the following sense: there are a permutation σ of J and $$ -automorphisms $\theta_j : M_p \rightarrow M_p$ such that α sends the j -th tensor factor onto the $\sigma(j)$ -th tensor factor via θ_j . Then*

$$h_c(\alpha) = \begin{cases} 0, & \text{if every orbit of } \sigma \text{ is finite,} \\ \infty, & \text{if } \sigma \text{ has an infinite orbit.} \end{cases}$$

In particular, no such product-type automorphism has finite nonzero CA entropy.

Proof. First suppose every orbit of σ is finite. Let $F \subset J$ be a finite union of σ -orbits. The finite tensor product

$$A_F = \bigotimes_{j \in F} M_p$$

is finite-dimensional and α -invariant. The union of these A_F ’s, over finite unions of orbits, is dense in A . On a finite-dimensional invariant subspace the orbit of each finite set admits a fixed finite-rank contractive approximation dimension independent of the orbit length, so the corresponding CA entropy is zero. Proposition 3.1(ii) from Kerr–Li therefore gives $h_c(\alpha) = 0$.

Now suppose σ has an infinite orbit. Choose $j_0 \in J$ whose forward iterates $\sigma^k(j_0)$, $k \geq 0$, are distinct. Inside the j_0 -th copy of M_p , identify an M_2 corner and let

$$u = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad v = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The real span $E = \text{span}_{\mathbb{R}}\{u, v\}$ is isometric to ℓ_2^2 on self-adjoint elements: for real a, b , $\|au + bv\| = (a^2 + b^2)^{1/2}$. Let Ω be any finite subset of the unit circle of E .

For distinct $x, y \in \Omega$, the Euclidean identity gives $\|x + y\| < 2$. We claim that Ω satisfies the additivity hypothesis of Kerr–Li Lemma 9.1. Given $x_0, \dots, x_{n-1} \in \Omega$, the element $\alpha^k(x_k)$ lies in the tensor factor indexed by $\sigma^k(j_0)$, and these tensor factors are distinct. Each $\alpha^k(x_k)$ is self-adjoint

of norm one. For each k , choose a state φ_k on that matrix factor with $\varphi_k(\alpha^k(x_k)) = 1$, and take arbitrary states on the remaining factors. The resulting product state φ satisfies

$$\varphi \left(\sum_{k=0}^{n-1} \alpha^k(x_k) \right) = n.$$

The triangle inequality gives the reverse upper bound, hence

$$\left\| \sum_{k=0}^{n-1} \alpha^k(x_k) \right\| = n.$$

Lemma 9.1 now yields $h_c(\alpha, \Omega) \geq \log |\Omega|$. Since finite subsets Ω of the unit circle can have arbitrarily large cardinality, $h_c(\alpha) = \infty$. $\square$

Limitations and Novelty Check

This is not a full answer to Question 9.4. It handles the natural permutation-product automorphisms of countable UHF tensor products, including the source paper’s tensor shift, but it does not address general AF automorphisms which mix Bratteli levels or act through more subtle approximately inner mechanisms.

The run’s lightweight indexes were searched for arXiv:0407386, the paper title, ‘CA entropy’, ‘simple AF algebra’, ‘finite nonzero CA entropy’, and related UHF/product-type keywords. No prior packet or attempt for this target was found. Web searches for exact CA-entropy/simple-AF phrases found the source paper and no later direct resolution during this pass.

References

References

- [1] D. Kerr and H. Li, *Dynamical entropy in Banach spaces*, Invent. Math. 162 (2005), 649–686; arXiv:math/0407386.

Human Review Recommendation

The main point for review is the scope of “product-type”. The proof is intended only for automorphisms induced by a tensor-factor permutation and local matrix-factor automorphisms. Check the finite-orbit zero-entropy reduction and the product-state norm-additivity argument in the infinite-orbit case.